EXERCISE

- Solve the following equations:

1. $(x^2D^2 - xD + 1)y = 0$
2. $(x^2D^2 + xD + 1)y = 0$
3. $(x^2D^2 + 1)y = 0,$
4. $(x^2D^2 + xD - 1)y = 4,$
5. $(x^2D^2 - 2)y = x^2 + \frac{1}{x},$
6. $(x^2D^2 - 3xD + 4)y = 2x^2,$
7. $(x^2D^2 - 2xD + 2)y = \frac{1}{x},$
8. $(x^2D^2 + xD - 1)y = \sin(\log x) + x\cos(\log x),$
9. $(x^2D^2 - 3xD + 4)y = 2x^2,$
10. $(x^2D^2 + xD - 4)y = x^2,$
11. $(x^2D^2 - 4xD + 6)y = \ln x^2,$
12. $(x^2D^2 - 3xD + 5)y = \sin(\log x),$
13. $(x^2D^2 - 3xD + 5)y = x^2\sin(\log x),$
14. $(x^2D^2 + xD + 1)y = \log x \sin(\log x),$
15. $(x^3D^3 + x^2D^2)y = x,$
16. $(x^3D^3 + 2x^2D^2 - xD + 1)y = \frac{1}{x}$
17. $(x^3D^3 + 2x^2D^2 + 2)y = 10\left(x + \frac{1}{x}\right),$
18. $(x^3D^3 - x^2D^2 + 2xD - 2)y = x^3,$
19. $(x^3D^2 + xD - 1)y = 3x^4,$
20. $(x^3D^3 + 2xD - 2)y = 3x + x^2\log x,$
21. $(x^3D^3 + 3x^2D^2 + xD + 8)y = 65\cos(\log x),$
22. $(x^3D^3 - 4x^2D^2 + 5xD - 2)y = x^3,$
23. $(x^3D^3 + 3x^2D^2 + xD + 1)y = x\log x,$
24. $(x^4D^4 + 6x^3D^3 + 4x^2D^2 - 2xD - 4)y = x^2 + 2\cos(\log x),$
25. $(x^4D^4 + 6x^3D^3 + 9x^2D^2 + 3xD + 1)y = (1 + \log x)^2,$
26. $(x + a)^2D^2y - 4(x + a)Dy + 6y = x,$
27. $(1 + x)^2D^2y + (1 + x)Dy + y = 4\cos\ln(1 + x),$
28. $(1 + x)^2D^2y + (1 + x)Dy + y = 2\sin\ln(1 + x),$
29. $(1 + x)^2D^2y - 3(1 + x)Dy + 4y = x,$
30. $(1 + x)^2D^2y + (1 + x)Dy = 4x^2 + 14x + 12,$
31. $(2x - 1)^3D^3y + (2x - 1)Dy - 2y = 0,$
32. $(2 + 3x)^2D^2y + 5(2 + 3x)Dy - 3y = x^2 + x + 1,$
33. $(2 + x)^2D^2y + (2 + x)Dy + 4y = 2\sin\{2\ln(2 + x)\}.$

• Solve the following equations using method of variation of parameter:

34. $x^2y''+xy'-y=x^2e^x$,

35. $x^2y''+xy'-y=x^2e^{2x}$,

36. $x^2y''+xy'-y=x$,

37. $x^2y''-xy'=x^3e^x$,

38. $x^2y''+3xy'+y=\dfrac{1}{(1-x)^2}$,

39. $x^2y''+xy'-y=x^2\log x$,

40. $x^2y''+2xy'=\log x$,

41. $x^2y''-2xy'+2y=x$,

42. $x^2y''+5xy'+4y=x\log x$,

43. $x^2y''-xy'+2y=x\log x$.

ANSWERS

1. $(c_1+c_2\ln x)x$,

2. $c_1\cos(\ln x)+c_2\sin(\ln x)$,

3. $\sqrt{x}\left[c_1\cos\dfrac{\sqrt{3}\ln x}{2}+c_2\sin\dfrac{\sqrt{3}\ln x}{2}\right]$,

4. $c_1x+\dfrac{c_2}{x}-4$,

5. $c_1x^2+\dfrac{c_2}{x}+\dfrac{1}{3}\left(x^2-\dfrac{1}{x}\right)\ln x$,

6. $(c_1+c_2\ln x)x^2+(x\ln x)^2$,

7. $c_1x+c_2x^2+\dfrac{1}{6x}$,

8. $c_1x+\dfrac{c_2}{x}-\dfrac{1}{2}\sin(\ln x)+\dfrac{x}{5}[2\sin(\ln x)-\cos(\ln x)]$,

9. $(c_1+c_2\ln x)x^2+x^2(\ln x)^2$,

10. $c_1x^2+c_2x^{-2}+\dfrac{1}{4}x^2\ln x$,

11. $c_1x^3+c_2x^2+\dfrac{1}{18}(3\ln x^2+5)$,

12. $x^2[c_1\cos(\ln x)+c_2\sin(\ln x)]+\dfrac{1}{8}[\sin(\ln x)+\cos(\ln x)]$,

13. $x^2\,[A\cos(\ln x)+B\sin(\ln x)]-\dfrac{x^2}{2}\ln x\cos(\ln x)$,

14. $c_1\cos(\ln x)+c_2\sin(\ln x)+\dfrac{\ln x}{4}[\sin(\ln x)-\ln x\cos(\ln x)]$,

15. $c_1+(c_2+c_3\ln x)x+\dfrac{x}{2}(\ln x)^2$,

16. $(c_1+c_2\ln x)x+\dfrac{c^3}{x}+\dfrac{1}{4x}\ln x$,

17. $\dfrac{c_1}{x}+x[c_2\cos(\ln x)+c_3\sin(\ln x)]+\dfrac{2}{x}\ln x+5x$,

18. $(c_1+c_2\ln x)x+c_3x^2+\dfrac{x^3}{4}$,

19. $c_1x+c_2x\ln x+c_3x(\ln x)^2+\dfrac{x^4}{9}$,

20. $x[c_1 + c_2 \cos(\ln x) + c_3 \sin(\ln x) + 3\ln x] + \frac{1}{2}x^2(\ln x - 2)$,

21. $\frac{c_1}{x^2} + x\{c_2 \cos(\sqrt{3}\ln x) + c_3 \sin(\sqrt{3}\ln x)\} + 8\cos(\ln x) - \sin(\ln x)$,

22. $c_1 x^2 + x^{\frac{5}{2}}\left\{c_2 x^{\frac{\sqrt{21}}{2}} + c_3 x^{-\frac{\sqrt{21}}{2}}\right\} - \frac{1}{5}x^3$,

23. $\frac{c_1}{x} + \sqrt{x}\left[c_2 \cos\left(\frac{\sqrt{3}\ln x}{2}\right) + c_3 \sin\left(\frac{\sqrt{3}\ln x}{2}\right)\right] + \frac{x}{2}\left(\ln x - \frac{3}{2}\right)$,

24. $c_1 x^2 + c_2 x^{-2} + c_3 \cos(\ln x) + c_4 \sin(\ln x) + \frac{1}{20}x^2 \ln x - \frac{1}{5}\ln x \sin(\ln x)$,

25. $(c_1 + c_2 \ln x)\sin(\ln x) + (c_3 + c_4 \ln x)\cos(\ln x) + (\ln x)^2 + 2\ln x - 3$,

26. $c_1(x+a)^2 + c_2(x+a)^3 + \frac{(x+a)}{2} - \frac{a}{6}$,

27. $c_1 \cos\{\ln(1+x) - c_2\} + 2\ln(1+x)\sin\ln(1+x)$,

28. $c_1 \cos\ln(1+x) + c_2 \sin\ln(1+x) - \ln(1+x)\cos\ln(1+x)$,

29. $(1+x)^2[c_1 + c_2 \ln(1+x)] + x + \frac{3}{4}$,

30. $c_1 + c_2 \ln(1+x) + \{\ln(1+x)\}^2 + x^2 + 8x$,

31. $c_1(2x-1) + c_2(2x-1)\cosh\left[\frac{\sqrt{3}}{2}\ln(2x-1) + c_3\right]$,

32. $c_1(3x+2)^{-1} + c_2(3x+2)^{\frac{1}{3}} + \frac{1}{405}(3x+2)^2 - \frac{1}{108}(3x+2) - \frac{7}{27}$,

33. $A\cos\{2\ln(2+x)\} + B\sin\{2\ln(2+x)\} - \frac{1}{2}\ln(2+x)\cos\{2\ln(2+x)\}$,

34. $ax + \frac{b}{x} + e^x\left(1 - \frac{1}{x}\right)$,

35. $ax + \frac{b}{x} + \frac{1}{8}e^{2x}\left(2 - \frac{1}{x}\right)$,

36. $ax + \frac{b}{x} + \frac{x^3}{8}$,

37. $a + bx + (x-1)e^x$,

38. $\frac{a}{x} + \frac{b}{x}\ln x + \frac{1}{x}\ln\left(\frac{x}{1-x}\right)$,

39. $ax + \frac{b}{x} + \frac{x^2}{3}\ln x - \frac{4x^2}{9}$,

40. $a + \frac{b}{x} + \frac{1}{2}(\ln x)^2 - \ln x$,

41. $ax + bx^2 - x(1 + \ln x)$,

42. $(a + b\ln x)x^{-2} + \frac{x}{27}(3\ln x - 2)$,

43. $x[a\cos(\ln x) + b\sin(\ln x)] + x\ln x$.